

A Self-Sustaining Theory of Epistemology Revisited

A Reduced Dynamic Model of Capacity, Defense, and Mobility

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Abstract

We model an epistemic institution as an open dynamical system in which workers produce information, part of which is validated into knowledge, and the rest imposes a processing burden. Using standard results from constrained optimisation, positive systems, and information theory, we derive (i) an allocation rule in which workers are employed only where net epistemic productivity exceeds the shadow price of capacity, (ii) a loop-gain condition separating institutions that reproduce their own capacity from those that depend on external entry, including a tipping point below which the institution collapses, (iii) a lower bound on epistemic diversity that is positive only when external entry is present, and (iv) a defense margin. Non-Immunity is stated as a design requirement, not a theorem.

The paper ends with “The End”

1 Information, knowledge, and burden

Epistemologist $i = 1, \dots, N$ employs $n_i \geq 0$ workers who produce gross information $q_i(n_i) = \alpha_i n_i^{\beta_i}$, with $\alpha_i > 0$ and $0 < \beta_i < 1$, so that returns are diminishing. A fraction $\theta_i \in [0, 1]$ is validated into knowledge. Per-period validated output and unvalidated information are

$$\Delta K(n) = \sum_i \theta_i q_i(n_i), \quad (1)$$

$$U(n) = \sum_i (1 - \theta_i) q_i(n_i). \quad (2)$$

We write U for unvalidated information. The earlier paper called it propaganda, but the definition is operational (a claim not yet validated), and includes honest conjectures, so a neutral name avoids implying intent [1, 3]. Unvalidated information is not useless, since discovery precedes validation, but it must be processed, and attention is scarce [5]. We model the burden as $B(U) = bU^\gamma$, $b > 0$, $\gamma > 1$, so that $B' > 0$ and $B'' > 0$.

2 Worker allocation

Let $c_i > 0$ be the marginal cost of a worker and Z the processing capacity, with the constraint $\sum_i q_i \leq Z$. The institution maximises

$$\Pi(n) = \Delta K(n) - B(U(n)) - \sum_i c_i n_i. \quad (3)$$

Define effective productivity $\psi_i(U) = \theta_i - (1 - \theta_i)B'(U)$, and let $\mu \geq 0$ be the multiplier on the capacity constraint.

Proposition 1 (Allocation). *The Karush–Kuhn–Tucker conditions [9] are*

$$\alpha_i \beta_i n_i^{\beta_i - 1} (\psi_i(U) - \mu) \leq c_i, \quad \text{with equality if } n_i > 0. \quad (4)$$

1. If $\psi_i(U) \leq \mu$, then $n_i = 0$: an epistemologist whose net productivity does not exceed the shadow price of capacity employs no workers.
2. With $\mu = 0$, $\psi_i > 0$ iff $U < U_i^\dagger$, where $U_i^\dagger = (\theta_i / ((1 - \theta_i)b\gamma))^{1/(\gamma-1)}$, which increases with θ_i .
3. The Hessian of the Lagrangian is $\text{diag}((\psi_i - \mu)q_i'') - B''(U)vv^\top$ with $v_i = (1 - \theta_i)q_i'$. It is negative definite wherever $\psi_i > \mu$ for all i , so a solution of (4) in that region is a strict local maximum.

Proof. Differentiating Π gives $\partial\Pi/\partial n_i = q_i' \psi_i - c_i$. The second derivatives are $\delta_{ij}\psi_i q_i'' - B''v_i v_j$, because the δ_{ij} terms combine as $\theta_i q_i'' - B'(1 - \theta_i)q_i''$. Since $q_i'' < 0$ and $B'' > 0$, both pieces are negative where $\psi_i > \mu$. The threshold in part 2 follows from $B'(U) = bU^{\gamma-1}\gamma$ set equal to $\theta_i/(1 - \theta_i)$. $\square$

Note that Π is not globally concave, since a convex B composed with a concave U need not be convex, so only the local statement holds.

3 Reproduction of capacity

Let K be the knowledge stock, Z capacity, and let $\delta_K, \delta_Z \in (0, 1]$ be depreciation rates. Take φ and r increasing, concave, C^1 , with $\varphi(0) = r(0) = 0$: validated output depends on capacity, and reproduction of capacity (training, memory) depends on knowledge. Let $\epsilon \geq 0$ be capacity contributed by external entrants net of integration cost, and $\Omega = B(U)$ the burden flow. The reduced system is

$$K_{t+1} = (1 - \delta_K)K_t + \varphi(Z_t), \quad (5)$$

$$Z_{t+1} = \max\{0, (1 - \delta_Z)Z_t + r(K_t) + \epsilon - \Omega\}. \quad (6)$$

Write $\epsilon_{\text{net}} = \epsilon - \Omega$ and define the *reproduction gain*

$$\Theta = \frac{\varphi'(0)r'(0)}{\delta_K \delta_Z}. \quad (7)$$

Proposition 2 (Self-sustainability and tipping). *Equilibria of (5)–(6) are the roots of $g(Z) = r(\varphi(Z)/\delta_K) + \epsilon_{\text{net}} - \delta_Z Z$, which is concave. At an equilibrium the Jacobian is nonnegative, with Perron root below 1 iff $\delta_K \delta_Z > \varphi'(Z^*)r'(K^*)$, equivalently $g'(Z^*) < 0$.*

1. If $\epsilon_{\text{net}} > 0$, there is a unique positive equilibrium and it is locally asymptotically stable.
2. If $\epsilon_{\text{net}} = 0$, the origin is an equilibrium, stable iff $\Theta < 1$. If $\Theta > 1$, there is also a unique positive stable equilibrium.
3. If $\epsilon_{\text{net}} < 0$, a positive equilibrium requires $\Theta > 1$. Then there are at most two positive roots: the smaller is unstable and acts as a tipping point (capacity below it collapses to 0), and the larger is stable. If $\max g < 0$, every trajectory collapses.

Proof. For a nonnegative Jacobian J , $\rho(J) < 1$ iff $I - J$ is a nonsingular M-matrix [10]; for the 2×2 case this is $\delta_K \delta_Z - \varphi' r' > 0$ [11]. Because $r \circ \varphi$ is concave, g is concave, so its roots are ordered with g' positive at the smaller and negative at the larger. Since $g'(0) = \delta_Z(\Theta - 1)$, a concave g with $g(0) < 0$ can reach a positive maximum only if $g'(0) > 0$, which gives part 3. $\square$

Thus $\Theta > 1$ is the precise sense in which an institution is *self-sustaining*: it can maintain capacity through its own reproduction loop. If $\Theta < 1$, positive capacity requires $\epsilon_{\text{net}} > 0$, that is, external entry that outweighs the burden.

4 Defense and attack

Let $A \geq 0$ be attack intensity, injecting unvalidated items so that the burden becomes $B(U + A)$. Since $B'' > 0$, $\partial^2 B / \partial U \partial A > 0$: the same attack is more damaging when the institution is already loaded, which is the flooding mechanism. Defensive capacity follows $Y_{t+1} = (1 - \delta_Y)Y_t + I_Y$, so $Y^* = I_Y / \delta_Y$. The defense margin is $Y^* - A$, and defense is viable iff $Y^* \geq A$. Defense here is methodological (proof, replication, source criticism, adversarial review), as in [1].

5 Diversity, entry, and exit

Let H_t be the distribution of hypotheses and methods, and D Shannon entropy [7]. Internal reproduction maps H_t to $T(H_t)$; we assume it may contract diversity by at most a factor $1 - \lambda$, meaning $D(T(H)) \geq (1 - \lambda)D(H)$. External entry mixes in an independent distribution G with $D(G) \geq D_G$:

$$H_{t+1} = (1 - \omega)T(H_t) + \omega G, \quad 0 < \omega < 1. \quad (8)$$

Proposition 3 (Diversity floor). *Entropy is concave [8], so $D(H_{t+1}) \geq (1 - \omega)(1 - \lambda)D(H_t) + \omega D_G$. Therefore*

$$\liminf_{t \rightarrow \infty} D(H_t) \geq \frac{\omega D_G}{1 - (1 - \omega)(1 - \lambda)}. \quad (9)$$

With $\omega = 0$ this is only $D(H_t) \geq (1 - \lambda)^t D(H_0) \rightarrow 0$: an echo chamber is possible.

This is a lower bound: entry makes a floor possible but does not guarantee high diversity, and it requires G to be genuinely independent ($D_G > 0$). Exit and re-entry preserve this independence and give departing members the option emphasised in [6]. Membership obeys the accounting identity $N_{t+1} = N_t + R_t + V_t + V_t^R - X_t$ (reproduction, entry, re-entry, exit). With proportional rates $R_t = rN_t$ and $X_t = xN_t$, and V fixed, $N^* = V/(x - r)$ when $x > r$. The condition $r \geq x$ means the institution replaces itself without entry, mirroring $\Theta > 1$.

6 Viability and Non-Immunity

Collecting the results, the institution is *viable* on a horizon if

$$\Theta > 1 \text{ or } \epsilon_{\text{net}} > 0, \quad Y^* \geq A, \quad \omega > 0, \quad (10)$$

together with a trajectory that stays above any tipping point from Proposition 2. The *Non-Immunity Principle* is a design requirement: the set $\mathcal{A}_{\text{int}}$ of methods allowed against the institution's own claims and procedures must contain the set $\mathcal{A}_{\text{ext}}$ used against external claims. It is not derived from the model, but it is what keeps the error-correction terms in the model operative, in line with [1, 3, 4, 2].

7 What was removed

The interdisciplinary table, the accounting and welfare sections, the network-science and AI sections, and the 15-step protocol were removed: they were heuristics without a model attached. The original Self-Sustainability Theorem (capacity is non-decreasing if net inflow is non-negative) was a tautology and is replaced by Proposition 2.

References

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